ELAMATHY SIVAKUMAR

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SUMMARY

Software engineer with 2.5+ years of experience building **Java** and **Spring Boot** microservices, RESTful APIs, and data integration workflows for a global bank, delivering 10+ features with zero critical defects and 90%+ JUnit coverage. Strong in **SQL** and Oracle schema tuning (~35% throughput gain), with recent **AI/ML** work in multi-agent systems and RAG. M.S. CS, GPA 3.9.

TECHNICAL SKILLS

Languages and Frontend: Java, Python, JavaScript, SQL, Angular, HTML5, CSS3, Bootstrap
Backend: Spring Boot, Spring MVC, JDBC, RESTful APIs, Microservices, FastAPI, Multithreading, Concurrency
Cloud: AWS (S3, Lambda)
Databases: Oracle, MySQL, MongoDB, ChromaDB, Stored Procedures, Query Optimization
AI/ML: LangChain, Model Context Protocol (MCP), RAG Pipelines, Multi-agent Systems, Ollama
Practices and Tools: Git, JUnit, Agile/Scrum, Code Review, Production Support, Full SDLC, AI-assisted Development

WORK EXPERIENCE

Graduate Teaching Assistant, Montclair State University **Sep 2025 to May 2026**

- Mentored **50+ students** through weekly office hours, lab sessions, and midterm and exam **review sessions** in Java, data structures, and OOP.
- Ran code reviews on student submissions covering **Java**, OOP, and backend design, helping students debug and refactor.
- Graded **100+ assignments** per semester with detailed feedback on code structure, naming, and testability, and helped the instructor manage the **Canvas** course site.

Software Engineer, Citibank N.A. (via LTIMindTree) **Aug 2021 to Oct 2023**

- Shipped **10+ Java and Spring Boot features** in a multi-engineer team across 3 time zones, with **zero critical defects** over 2+ years.
- Designed and owned **10+ RESTful APIs and microservices** with JDBC and **AWS (S3, Lambda)** that validated and integrated partner-team data feeds with our own SQL schemas, using multithreading and concurrency, with **90%+ JUnit coverage**.
- Improved throughput **~35%** by redesigning Oracle schemas with targeted indexes, query tuning, and stored procedures.
- Built an **Angular dashboard proof of concept** and worked with product managers and business stakeholders in sprint reviews, turning workflow pain points into backend and product improvements.
- Provided **production support** and applied secure coding practices, including input validation, role-based access control, and data encryption, to protect sensitive financial data in line with internal security requirements.
- Mentored junior engineers through pairing and code reviews, and took part in sprint planning, stand-ups, and retros.

Software Engineer, Cognizant **Mar 2021 to Jun 2021**

- Built web interfaces in JavaScript, HTML5, and CSS3, and contributed to **Java and Spring** feature development and testing, writing **SQL** queries against the application database.
- Took part in sprint planning, daily stand-ups, and retrospectives, following the **Agile SDLC** from requirements to deployment.
- Used **Git** version control with pull-request workflows and peer code reviews in a professional team setting.

PROJECTS

MCP-Based Multi-Agent AI Framework, Python, FastAPI, ChromaDB github.com/Elamathy-S/multi-agent-ai-framework

- Architected a multi-agent system that simulates financial workflows, coordinating **4 specialist agents** (customer profiling, credit scoring, risk assessment, portfolio analysis) via MCP and JSON-RPC 2.0, with a FastAPI backend and SSE streaming.
- Built a **RAG pipeline** with ChromaDB to ground agent reasoning in domain data, using a local LLM (Ollama, phi3).
- Raised output reliability from **60% to 90%** with structured validation checks and iterative prompt engineering, cutting manual review effort by **~40%** across simulated financial decisioning workflows.
- Built an orchestrator to coordinate multi-step workflows across agents, with structured logging and full traceability.

EDUCATION

M.S. Computer Science, Montclair State University, GPA 3.9/4.0 **May 2026**

B.E. Computer Science and Engineering, Anna University, GPA 3.7/4.0 **May 2021**

CERTIFICATIONS

Oracle Certified Professional, Java SE 11 Developer